MILITARY SPECIFICATION

SURVEYING SET, ASTRONOMIC POSITION, PACKAGING OF

1. SCOPE

1.1 This specification covers the requirements for preservation, packaging, packing and marking of the astronomic position surveying set, (SC 6675-97-CL-E34).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-S-781	- Strapping, Steel, Flat and Seals.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Box, Fiberboard.
PPP-C-1120	- Cushioning Material, Uncompressed Bound Fiber for Packaging.
PPP-P-40	- Packaging and Packing of Hand Tools.

Military

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation, Methods of.
MIL-S-3534	- Surveying Instruments and Accessories, Packaging of.

FSC 6675

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MIL-C-3774	- Crates, Wood; Open, 12,000- and 16,000-Pound Capacity.
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservative and Break-In.
MIL-G-28554	- Generator Sets, Mobile Electric Power; Packaging of.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

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3. REQUIREMENTS

3.1 Preproduction pack. The supplier shall furnish a preproduction pack for examination and test within the time frame specified to prove prior to starting production packaging and packing, that the applied preservation, packaging, packing and marking comply with the requirements of this specification. Examination and test shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.2 Materials. Materials shall be as specified herein and in the applicable specifications and standards. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.3 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

3.3.1 Level A.

3.3.1.1 Methods of preservation. Methods of preservation specified herein shall conform to the requirements of MIL-P-116.

3.3.1.2 Altimeter-barometer, level, plumb bobs and sheaths, target set, theodolites and tripod. These items shall be preserved and packaged in accordance with the requirements of MIL-S-3534, level A.

3.3.1.3 Ax, die set, drills, files, hammers, knife, machettes and sheaths, mattock, pliers, saw, screwdriver, shovel, measuring tapes and wrenches. These items shall be preserved and packaged in accordance with the requirements of PPP-P-40, level A.

3.3.1.4 Cotton duck bag, ballpoint pens, books, charts, clipboard files, envelopes, flashlights, printed forms, ink cartridges, insulating tape, signal lights, writing pads, padlocks, lens paper, inking pens, pencils, pencil pointers, drafting scale, can spout, soldering iron, and thermometer. These items shall be preserved by method IC-1. Items of like description shall be preserved together in quantities required for each set when item configuration of size permits, otherwise items shall be individually preserved.

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3.3.1.5 Baseplates, calculating machine, clamp handle, chart take-up drive, compasses, antenna mast, multimeter, nails, tent pins, solder, tents, tension handles, and trivets. Each item shall be individually preserved by method III. Containers shall be close-fitting boxes conforming to PPP-B-636, V3c, W5c, or W6c as applicable, style optional. Cushioning and blocking shall be provided as required to prevent or control movement of items within the boxes. Items shall be collapsed or folded when applicable to provide a box of minimum cube. After closure the boxes shall be waterproof sealed with tape as specified for slotted style boxes in the appendix to the box specification.

3.3.1.6 Binocular, oscillograph, radio receiver, and recorder. Each item shall be individually preserved according to method IIb. When a case is furnished, the case shall be the inner container, otherwise the inner container shall be a close-fitting box conforming to PPP-B-636, type CF, class domestic, variety SW, grade 125. The outer container shall be a box conforming to PPP-B-636, W5c, style optional. The outer container shall be waterproof sealed with tape as specified for slotted style boxes in the appendix to the box specification.

3.3.1.7 Chronometers. The balance mechanism of each chronometer shall be blocked following procedures recommended by the supplier. Each chronometer shall then be placed in the case provided and preserved according to method III. The container shall be a box conforming to PPP-B-636, W5c, style optional. A minimum of 3-1/2 inches of cushioning material conforming to PPP-C-1120, type III shall be provided between all surfaces of the chronometer case and the box. After closure the box shall be waterproof sealed with tape as specified for slotted style boxes in the appendix to the box specification.

3.3.1.8 Insulator spreaders. Insulator spreaders of like description shall be preserved together according to method III in a paperboard box conforming to PPP-B-566.

3.3.1.9 Electrical wire and cable. Each coil or spool of wire or cable shall be preserved according to method III in close-fitting boxes conforming to PPP-B-636 W6c, style optional.

3.3.1.10 Gasoline can. Each can shall be flushed or fogged with preservative lubricating oil conforming to MIL-L-21260, type I, grade 30. Excess preservative shall be drained.

3.3.1.11 Cement and plaster. Bags of cement and cans of plaster shall not require unit packaging.

3.3.1.12 Water can. The water can shall not require preservation or unit packaging.

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3.3.1.13 Softwood lumber and plywood. The 2-inch by 4-inch lumber shall be strapped together and the plywood sheets shall be strapped together in quantities specified for each set. Edge protectors shall be provided to prevent damage to the edges. Strapping shall conform to QQ-S-781, type I, class B, .023 thick by 1/2 inch wide.

3.3.1.14 Generator set (MAP). The generator set shall be preserved, packaged, and packed in accordance with MIL-G-28554, level A except that strapping shall not be required.

3.3.1.15 Intermediate and consolidated packaging. Bags of cement and cans of plaster shall be intermediate packaged, in quantities specified for each set, in close-fitting boxes conforming to PPP-B-621, class 1, style optional or PPP-B-601, domestic type, style optional. Other components shall be consolidated packaged in the chests where space permits and any remaining components, excluding the lumber and plywood, shall be consolidated packaged in close-fitting boxes conforming to PPP-B-621, class 1, style optional or PPP-B-601, domestic type, style optional. Cushioning and blocking shall be provided to control or prevent movement of components within the chests or boxes. Strapping of boxes shall not be required.

3.3.2 Level C. The components of each complete surveying set shall be preserved and packaged to afford protection against deterioration and damage during shipment and handling from the supplier to the initial destination for immediate use.

3.4 Packing. Packing shall be level A, B, or C as specified (see 6.2).

3.4.1 Level A. The components of each complete surveying set, packaged as specified in 3.3, shall be packed in a crate conforming to MIL-C-104, type I, class 1 or 2, style a. Blocking, bracing, and anchoring shall be in accordance with the appendix to the crate specification and in accordance with MIL-STD-1186. Strapping shall be zinc coated. Contents shall be arranged to provide a crate of minimum cube.

3.4.2 Level B. The components of each complete surveying set, packaged as specified in 3.3, shall be packed in a crate conforming to MIL-C-3774, type I. Blocking, bracing, anchoring, and waterproofing shall be in accordance with the appendix to the crate specification and in accordance with MIL-STD-1186. Strapping need not be zinc coated. Contents shall be arranged to provide a crate of minimum cube.

3.4.3 Level C. Each complete surveying set shall be packed to assure carrier acceptance and safe delivery to destination at the lowest rating in containers complying with Uniform Freight Classification rules or National Motor Freight Classification rules.

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3.5 Marking. Marking shall be in accordance with MIL-STD-129.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The supplier is responsible for insuring that supplies and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction pack inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction pack inspection.

4.3.1 Examination. The preproduction pack shall be examined for the defects listed in 4.4.4. Presence of one or more defects shall be cause for rejection of the preproduction pack.

4.3.2 Test. The preproduction pack shall be subjected to the railroad car guided impact test as specified in appendix A to MIL-STD-1186. The railcar shall strike a string of five empty cars with draft gear extended and the brakes set. The speed of the car at impact shall be not less than 10 miles per hour or not more than 11 miles per hour. Shifting of contents, loosening or breaking of anchoring, blocking, or bracing within the crate shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Inspection stages. Inspection shall be in three stages as follows:

(a) The first stage shall include inspection of procedures, methods, materials, containers, and packaging marking prior to placing the packages into consolidated containers.

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(b) The second stage shall include inspection of closure and marking of consolidated containers, and the arrangement and blocking and bracing of consolidated containers in the shipping containers.
(c) The third stage shall include inspection of the closed, strapped, and marked shipping containers as prepared for shipment.

4.4.2 Unit of product. For the purpose of inspection, all packages or packs prepared for any applicable inspection stage that will comprise a complete pack shall be considered a unit of product.

4.4.3 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.4 Examination. Samples selected in accordance with 4.4.3 shall be examined for the defects marked "X" for the applicable level in table I. Acceptable quality level (AQL) shall be 2.5 percent defective.

Table I. Classification of Defects

No.		Level A	Level B	Level C
101.	Materials, methods or containers not as specified. Each incorrect material, method or container shall constitute one defect.	X	X	-
102.	Components not preserved in accordance with referenced documents as specified.	X	-	-
103.	Boxes not waterproof sealed as specified.	X	-	-
104.	Balance mechanism of chronometers not blocked as specified.	X	-	-
105.	Cushioning not provided for chronometers as specified.	X	-	-
106.	Gasoline cans not flushed or fogged as specified.	X	-	-
107.	Edge protectors not provided for lumber and plywood as specified.	X	-	-
108.	Intermediate and consolidated packaging not as specified.	X	-	-
109.	Blocking, bracing, anchoring, or waterproofing not provided as specified.	X	X	-
110.	Strapping not zinc coated.	X	-	-
111.	Contents not arranged to provide crates of minimum cube.	X	X	-
112.	Marking illegible, incomplete, incorrect or missing.	X	X	X

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5. PREPARATION FOR DELIVERY

(This section not applicable to this specification.)

6. NOTES

6.1 Intended use. This specification is intended to be used for direct reference in contracts or purchase orders for the preservation, packaging, packing and marking of Surveying Set, Astronomic Position (FSN 6675-641-8897) for MAP (FSN 6675-926-1248).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Time frame required for submission of the preproduction pack (see 3.1).
(c) Level of preservation and packaging and level of packing required (see 3.3 and 3.4).

6.3 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to approval of the contracting officer. Approval of the preproduction pack will not relieve the supplier of his obligation to preserve, package, pack, and mark the set components in accordance with this specification.

Custodian:

Army - ME

Preparing activity:

Army - ME

Project No. 6675-A169

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-S-52729(ME) Surveying Set, Astronomic Position, Packaging Of

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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